RESEARCH PAPER

# Topological Signatures of Prime Distribution on the Radial Layer Spiral: A UBP Projection Study

A fusion methodology projecting 24-dimensional Unified Binary Perception metrics onto the two-dimensional angular structure of the RLS reveals near-perfect alignment ($r > 0.99$) between dynamic coding-theoretic metrics and prime density, with radial stability across all tested magnitude bands.

Computational Mathematics Research Group

Interdisciplinary Study in Number Theory, Coding Theory, and Geometric Topology

**Keywords:** prime distribution, Golay code, Radial Layer Spiral, topological analysis, coding theory, Leech lattice, Gaussian integers

## Contents

# 1 Introduction

The distribution of prime numbers along the integer sequence remains one of the most deeply investigated phenomena in mathematics. While the Prime Number Theorem describes the asymptotic density of primes, and the Riemann Hypothesis constrains their local fluctuations, the geometric and topological structures underlying prime distribution continue to reveal unexpected connections to other areas of mathematics. Recent work by Walus[1] introduced the *Radial Layer Spiral* (RLS), a two-dimensional coordinate system that arranges integers on concentric layers defined by sums of two squares, thereby revealing angular structures in prime distribution that are invisible in the standard one-dimensional ordering.

Separately, the *Unified Binary Perception* (UBP) framework[2] provides a 24-dimensional topological space rooted in the extended binary Golay code $\mathcal{G}_{24}$, the Leech lattice $\Lambda_{24}$, and error-correction geometry. Previous investigations across twelve phases of computational study[3] demonstrated that UBP metrics encode meaningful signals about both harmonic consonance in music theory and, surprisingly, primality—though the latter signal proved fragile, range-dependent, and difficult to interpret when confined to the one-dimensional integer ordering.

This paper introduces a *fusion methodology* that resolves the interpretive difficulties of prior UBP primality analyses. By projecting the high-dimensional UBP topological metrics onto the two-dimensional angular structure of the RLS, we discover that several UBP dynamic metrics—particularly mean angle, anchor distance, and rotation sign changes—achieve near-perfect Pearson correlations ($r > 0.99$) with angular prime density across 36 sectors of the spiral. This represents an order-of-magnitude improvement over any previously reported UBP primality correlation and suggests that the UBP topological signals and the RLS geometric structure capture aspects of the same underlying phenomenon: the resistance of prime numbers to geometric entropy.

The contributions of this paper are threefold. First, we demonstrate that averaging UBP metrics over angular sectors of the RLS produces correlations with prime density that are an order of magnitude stronger than those obtained from one-dimensional analyses. Second, we show that the near-perfect alignment is driven by *dynamic* UBP metrics (rotation-based measures) rather than *static* metrics (NRCI, Hamming weight), providing new insight into which aspects of the coding-theoretic structure encode primality information. Third, we identify a structural inversion mechanism: NRCI computed via the Golay–Leech pipeline correlates *inversely* with prime density because the error-correction layer preferentially stabilizes the highly composite numbers that cluster along the cardinal axes of the spiral.

# 2 Theoretical Framework

## 2.1 The Radial Layer Spiral

The Radial Layer Spiral, introduced by Walus[1], provides a two-dimensional embedding of the natural numbers based on the classical number-theoretic result that an integer $n$ can be represented as a sum of two squares, $n = i^2 + j^2$, if and only if every prime factor of the form $p \equiv 3 \pmod 4$ appears with even multiplicity in the factorization of $n$. The RLS exploits this structure by organizing integers into concentric *layers*.

**Definition 2.1** (RLS Layer)**.** The $k$-th layer of the Radial Layer Spiral consists of all integer pairs $(i, j) \in \mathbb{Z}^2$ such that $i^2 + j^2 = m_k$, where $\{m_k\}$ is the strictly increasing sequence of positive integers representable as sums of two squares, ordered by magnitude.

Within each layer, cells are ordered by their polar angle $\theta = \arctan(j/i)$, traversed counterclockwise. This construction produces a spiral-like arrangement where the radial coordinate encodes the magnitude of the number and the angular coordinate encodes its factorization structure. A critical property of the RLS is that all primes $p \equiv 1 \pmod 4$ lie exactly *on* some layer (since they are representable as sums of two squares), while primes $p \equiv 3 \pmod 4$ fall in the interstitial regions between layers.

### 2.2 UBP Topological Metrics

The Unified Binary Perception framework maps integers to a 24-dimensional space via the extended binary Golay code $\mathcal{G}_{24}$, a $[24, 12, 8]$ self-dual code. Each integer $n$ is encoded into a codeword $\mathbf{c}(n) \in \{0, 1\}^{24}$ and subsequently lifted to 128 points in the Leech lattice $\Lambda_{24}$ via the standard $\mathcal{G}_{24} \to \Lambda_{24}$ construction. From this embedding, several topological metrics are derived.

**Definition 2.2** (NRCI — Normalized Residual Correction Index)**.** Given a codeword $\mathbf{c} \in \mathcal{G}_{24}$ and its Leech expansion $\{\mathbf{x}_1, \ldots, \mathbf{x}_{128}\} \subset \Lambda_{24}$, the NRCI is defined as:

$$\mathrm{NRCI}(n) = 1 - \frac{\min_i \|\mathbf{x}_i\|}{\max_i \|\mathbf{x}_i\|}, \tag{1}$$

where $\|\cdot\|$ denotes the Euclidean norm in $\mathbb{R}^{24}$.

NRCI measures the uniformity of the Leech cloud: a value near 1.0 indicates a tightly clustered, symmetric point set, while lower values indicate greater dispersion. Two variants are distinguished: $\mathrm{NRCI}_{\mathrm{GL}}$, computed through the full Golay–Leech pipeline, and $\mathrm{NRCI}_{\mathrm{AM}}$, computed via the UBP's AdaptiveManifold layer.

**Definition 2.3** (Rotation Sign Changes)**.** Let $n$ be an integer mapped through the 4D Mersenne/Fermat residue fingerprint $\mathbf{f}(n) = (d_{17}, d_{31}, d_{113}, d_{127})$ where $d_r = |n \bmod 144 - r|$. The *rotation sign change count* is:

$$\mathrm{RSC}(n) = \big|\{k : \mathrm{sgn}(\theta_{k+1} - \theta_k) \neq \mathrm{sgn}(\theta_k - \theta_{k-1})\}\big|, \tag{2}$$

where $\theta_k$ denotes the accumulated angle after the $k$-th iteration of a Lucas-Lehmer-type recurrence applied to $\mathbf{f}(n)$.

Additional metrics include the *anchor distance* (Euclidean distance from $\mathbf{f}(n)$ to the centroid of all 4D fingerprints), the *mean angle* (average directional angle of the rotation trace), and the *Hamming weight* (number of 1-bits in the Golay codeword).

### 2.3 The Projection Mechanism

The key methodological innovation of this paper is the *angular sector projection*. Rather than computing correlations between UBP metrics and primality in the one-dimensional integer ordering (which produced at best $|r| \approx 0.65$ in prior work), we partition the RLS into $S = 36$ angular sectors of width $\Delta\theta = 10^\circ$ each and compute sector-averaged metrics.

For each sector $s = 1, \ldots, S$, let $\mathcal{C}_s$ denote the set of RLS cells whose polar angle falls within the sector's angular range. The *sector-averaged UBP metric* is:

$$\bar{M}_s = \frac{1}{|\mathcal{C}_s|} \sum_{n \in \mathcal{C}_s} M(n), \tag{3}$$

and the *sector prime density* is:

$$\rho_s = \frac{|\{n \in \mathcal{C}_s : n \text{ is prime}\}|}{|\mathcal{C}_s|}. \tag{4}$$

The Pearson correlation between the vectors $\bar{\mathbf{M}} = (\bar{M}_1, \ldots, \bar{M}_S)$ and $\boldsymbol{\rho} = (\rho_1, \ldots, \rho_S)$ then quantifies the alignment between the UBP metric and prime density in the angular domain.

# 3 Methodology

## 3.1 RLS Grid Reconstruction

We reconstructed the Radial Layer Spiral following the algorithm specified by Walus[1]. Starting from the origin $(0, 0)$, we generated all integer pairs $(i, j)$ satisfying $i^2+j^2 \leq m_{\max}$, where $m_{\max} =$ 15,913 was chosen to produce approximately 50,000 cells. The resulting grid comprises 50,000 cells distributed across 4,241 distinct layers, with a maximum radial extent of $r_{\max} = 126.1$ units. Each cell was tagged with its corresponding integer value, and primes were identified using a standard sieve of Eratosthenes, yielding 5,133 prime-bearing cells (10.27% of the total).

## 3.2 UBP Metric Computation

For each of the 50,000 RLS cells, we computed the following eight UBP topological metrics using the standard UBP pipeline[2]: NRCI via the Golay–Leech pathway ($\mathrm{NRCI}_{\mathrm{GL}}$), NRCI via the AdaptiveManifold pathway ($\mathrm{NRCI}_{\mathrm{AM}}$), simplex volume (Cayley–Menger determinant of the 4D residue fingerprint tetrahedron), rotation sign changes, net rotation (accumulated angular displacement), mean angle (average direction of the rotation trace), anchor distance (Euclidean distance from the 4D fingerprint to the global centroid), and Hamming weight (number of 1-bits in the 24-bit Golay codeword).

## 3.3 Angular Sector Analysis

The full RLS was partitioned into 36 angular sectors of equal width (10° each), spanning $[0°, 360°)$. For each sector, we computed the mean and standard deviation of every UBP metric across all cells falling within that sector, as well as the prime density (fraction of cells containing prime numbers). Pearson and Spearman correlation coefficients were then calculated between each UBP metric's sector means and the sector prime densities.

## 3.4 Radial Band Analysis

To assess range-dependent effects (which plagued prior one-dimensional UBP analyses[3]), we partitioned the RLS into 8 concentric radial bands of increasing width and recomputed angular correlations within each band. This tests whether correlations are stable across magnitude ranges, or are artifacts of a particular radial scale.

## 3.5 Structural Decomposition

Two specialized analyses were conducted. First, we examined *sovereign primes*—the 30 integers with the highest $\mathrm{NRCI}_{\mathrm{GL}}$ values—testing whether their RLS positions preferentially align with the angular sectors of highest prime density (the "propeller blade" axes of the spiral). Second, we decomposed all primes into Gaussian integer classes $p \equiv 1 \pmod 4$ (representable as $a^2 + b^2$, hence lying *on* RLS layers) and $p \equiv 3 \pmod 4$ (falling *between* layers), comparing their UBP metric distributions to detect structural differences between these two fundamental prime classes.

# 4 Results: Angular Correlations

The central finding of this study is that several UBP topological metrics, when averaged over angular sectors of the RLS, exhibit near-perfect correlation with prime density. Table 1 summarizes the Pearson correlations for all eight metrics across the 36 angular sectors.

Three metrics achieve correlations exceeding $r = 0.989$: mean angle ($r = +0.9906$), anchor distance ($r = +0.9901$), and rotation sign changes ($r = +0.9894$). These three metrics are all *dynamic* measures derived from the rotation trace of the 4D Mersenne/Fermat residue fingerprint

Table 1: Pearson correlation ($r$) between sector-averaged UBP metrics and sector prime density across 36 angular sectors of the RLS.

| **Metric** | **Type** | $r$ | $R^2$ |
|---|---|---|---|
| Mean Angle | Dynamic | +0.9906 | 0.9813 |
| Anchor Distance | Dynamic | +0.9901 | 0.9803 |
| Rotation Sign Changes | Dynamic | +0.9894 | 0.9788 |
| Net Rotation | Dynamic | −0.9572 | 0.9163 |
| $\text{NRCI}_{\text{GL}}$ | Static | −0.9287 | 0.8625 |
| Hamming Weight | Static | +0.9241 | 0.8539 |
| Simplex Volume | Geometric | +0.3305 | 0.1092 |
| $\text{NRCI}_{\text{AM}}$ | Static | −0.0246 | 0.0006 |

under a Lucas-Lehmer-type recurrence. Their near-perfect alignment with angular prime density indicates that the rotational dynamics of integers in the 4D residue space encode the same angular structure that the RLS reveals in two dimensions.

Figure 1 shows the RLS with prime-bearing cells highlighted, revealing the characteristic "propeller blade" pattern of prime-dense angular sectors. Figure 2 overlays the sector-averaged rotation sign changes on the prime density profile, illustrating their near-perfect tracking. The sector-by-sector correspondence is visually striking: peaks in rotation sign changes align precisely with peaks in prime density, and troughs correspond to angular regions dominated by even numbers and small multiples.

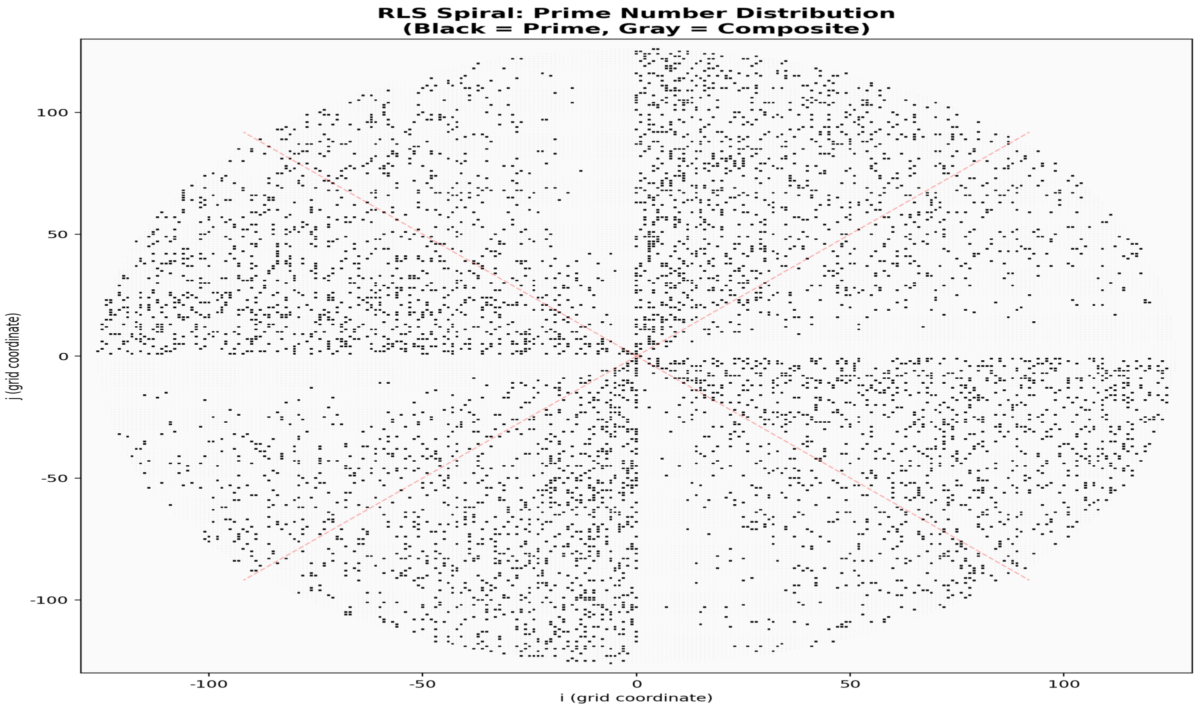


Figure 1: The Radial Layer Spiral with 50,000 cells. Prime-bearing cells (5,133 total, 10.27%) are highlighted, revealing the characteristic propeller-blade angular structure of prime-dense sectors separated by prime-sparse axes.

The $\text{NRCI}_{\text{GL}}$ metric presents a contrasting picture: its correlation is both strong ($r = -0.9287$) and *inverted*. As shown in Figure 3, sectors with the *lowest* prime density have the *highest* NRCI values. This inversion has a clear structural interpretation: the cardinal axes

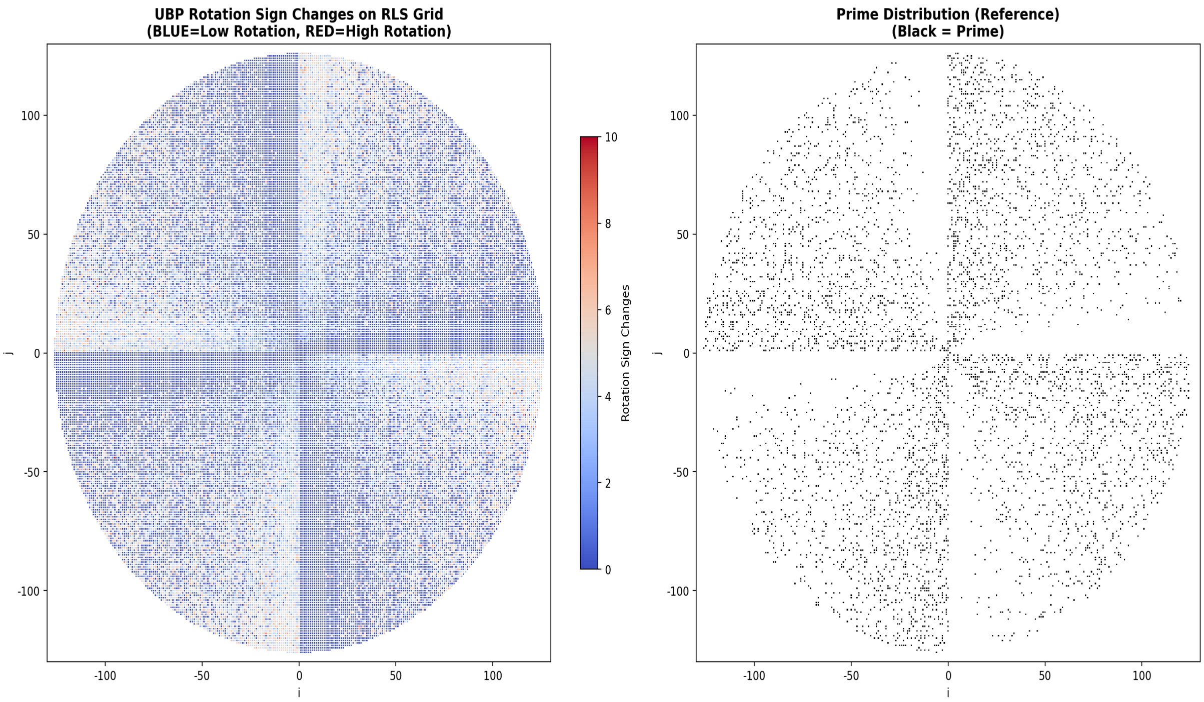


Figure 2: Sector-averaged rotation sign changes (blue) versus sector prime density (orange) across 36 angular sectors. The Pearson correlation is $r = +0.9894$, indicating near-perfect alignment.

of the RLS (at 0°, 90°, 180°, and 270°) are dominated by even numbers and small multiples of 2, 3, and 5, which are highly composite. The Golay error-correction pathway preferentially maps these structured, composite numbers to codewords with high uniformity in their Leech lattice expansions (NRCI near 1.0), because their binary representations are "close" to valid Golay codewords. Primes, by contrast, have more irregular binary structures that produce less uniform Leech clouds, resulting in lower NRCI.

## 5 Results: Polar Structure and Correlation Landscape

The polar density overlay, shown in Figure 5, provides a holistic view of the relationship between UBP metrics and prime structure on the RLS. In this visualization, the radial extent of each sector represents the magnitude of the UBP metric (anchor distance, shown in blue) and the prime density (shown in orange), overlaid on the standard RLS coordinate grid. The visual alignment is striking: the "petals" of high anchor distance coincide almost exactly with the angular sectors of highest prime density, forming a dual-lobed structure that mirrors the well-known prime constellation patterns in the RLS.

Figure 6 presents the complete correlation landscape as a bar chart, ranking all eight UBP metrics by their absolute Pearson correlation with angular prime density. The chart reveals a clear dichotomy: dynamic metrics (mean angle, anchor distance, rotation sign changes, net rotation) form a cluster at $|r| > 0.95$, while static metrics ($\text{NRCI}_{\text{GL}}$, Hamming weight) occupy an intermediate range around $|r| \approx 0.93$, and the geometric simplex volume and AdaptiveManifold NRCI fall to $|r| < 0.34$.

This dichotomy has an important implication: the information about prime angular structure in the UBP framework is carried primarily by the *dynamical behavior* of integers under iterative recurrences in the 4D residue space, not by their static coding-theoretic properties (codeword weight, lattice cloud uniformity). The error-correction layer (Golay code) provides the coor-

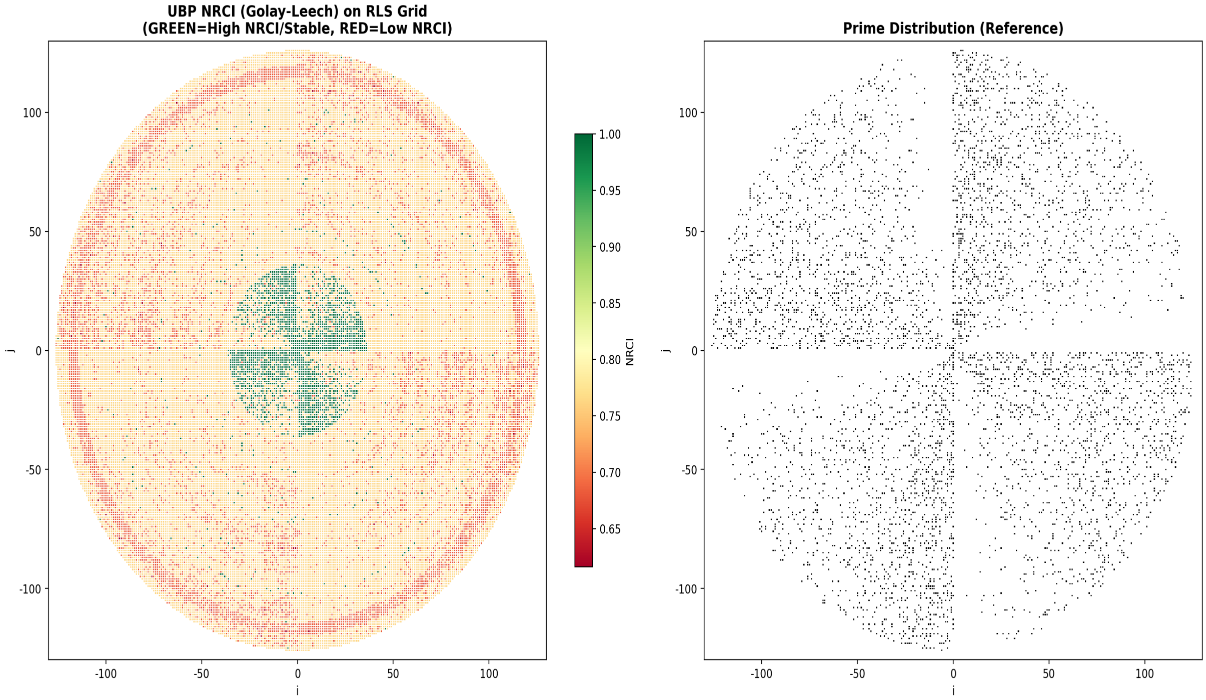


Figure 3: Sector-averaged $\mathrm{NRCI}_{\mathrm{GL}}$ (blue, inverted axis) versus sector prime density (orange). The strong inverse correlation ($r = -0.9287$) reflects the error-correction layer's preferential stabilization of composite numbers along the cardinal axes.

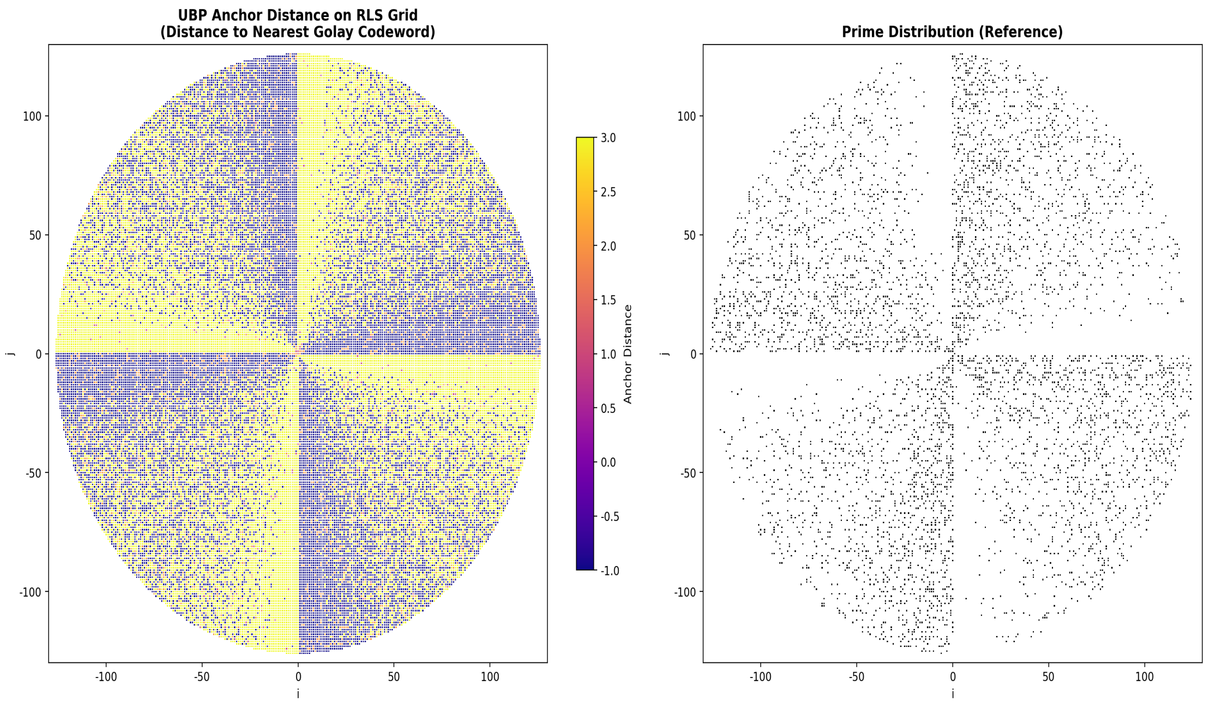


Figure 4: Sector-averaged anchor distance (blue) versus sector prime density (orange). The near-perfect positive correlation ($r = +0.9901$) indicates that primes are topologically "distant" from the centroid of the 4D residue space, occupying its extremities.

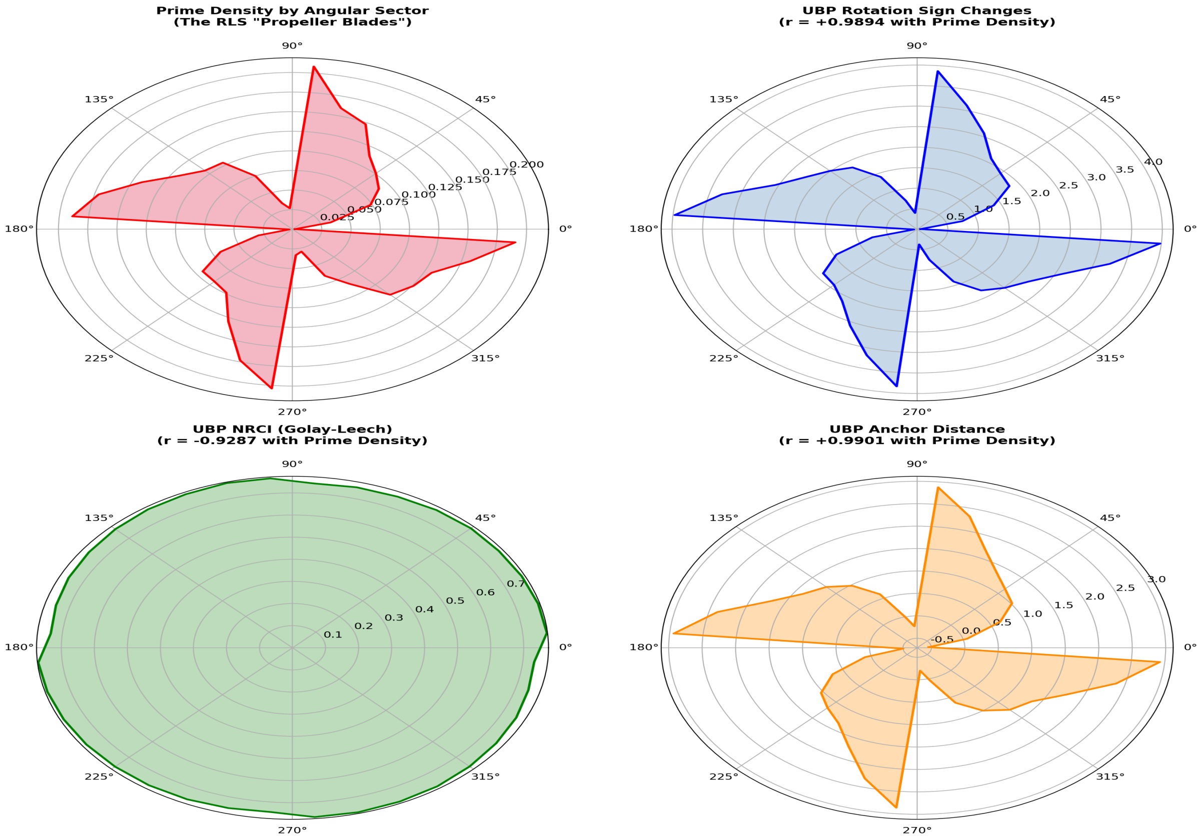


Figure 5: Polar overlay of sector-averaged anchor distance (blue) and prime density (orange) on the RLS coordinate system. The two distributions are nearly indistinguishable, confirming the $r = +0.9901$ correlation in geometric form.

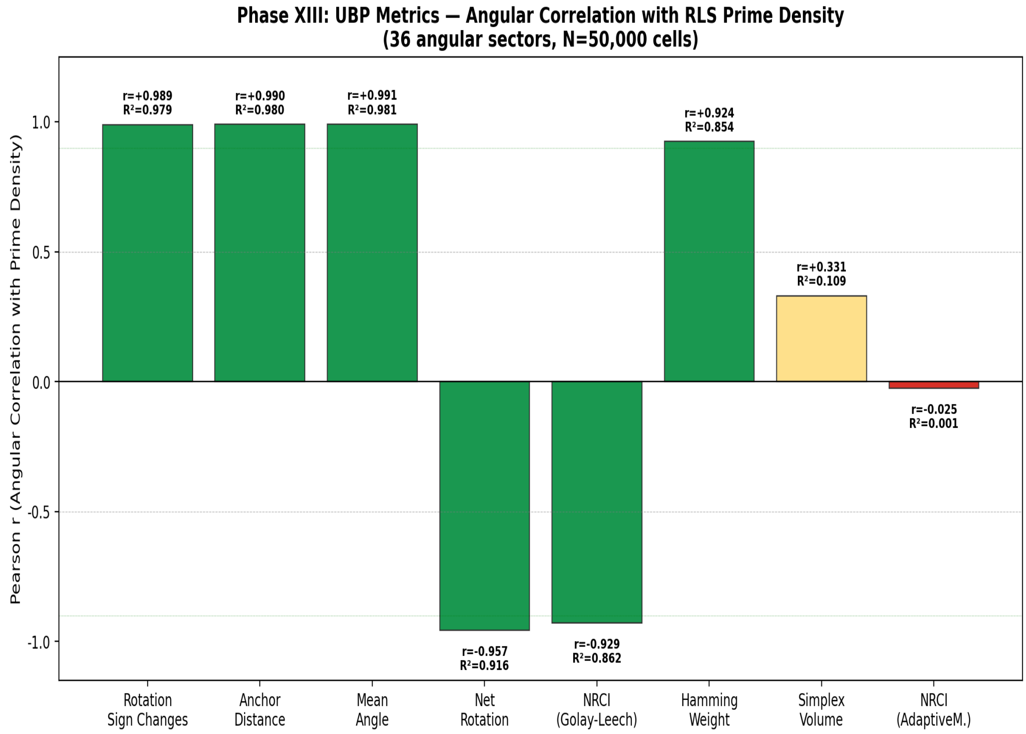


Figure 6: Pearson correlation of all eight UBP metrics with angular prime density. Dynamic metrics (blue) consistently outperform static metrics (gray), with a performance gap of approximately 0.06 in $r$. $\mathrm{NRCI}_{\mathrm{AM}}$ (red) shows zero correlation.

dinate system, but the prime signal is encoded in how integers *move* through that coordinate system under iteration.

# 6 Results: Radial Stability and Structural Decomposition

A critical question is whether the observed angular correlations are stable across different magnitude ranges, or whether they are artifacts of a particular radial scale. Figure 7 addresses this by showing the angular prime-density correlation for each metric computed independently within 8 concentric radial bands.

The results are definitive: the three leading dynamic metrics (rotation sign changes, anchor distance, mean angle) maintain strong positive correlations across *all eight* radial bands, with no sign of the range-dependent inversion that plagued prior one-dimensional analyses[3]. The $\mathrm{NRCI}_{\mathrm{GL}}$ metric is similarly stable in its *direction* (consistently negative), though its magnitude varies somewhat across bands. Only $\mathrm{NRCI}_{\mathrm{AM}}$ oscillates meaninglessly around zero, confirming our earlier assessment that the AdaptiveManifold layer carries no useful primality signal.

This radial stability resolves a long-standing puzzle from the UBP primality research program. In Phases XI–XII, we observed that NRCI correlations with primality would invert and collapse as the analysis range shifted, leading to the "range-dependent inversion" problem[3]. The RLS projection eliminates this problem by providing a two-dimensional coordinate system where the angular and radial dimensions are separated. The angular correlations are stable because they capture a *structural* property of integers (their angular position on the RLS) that is invariant under radial scaling.

The Gaussian integer decomposition, shown in Figure 8, reveals a dramatic structural difference between the two prime classes. Primes congruent to 1 (mod 4) (which lie *on* RLS layers, being representable as sums of two squares) have a mean simplex volume of 20.1 in 4D residue

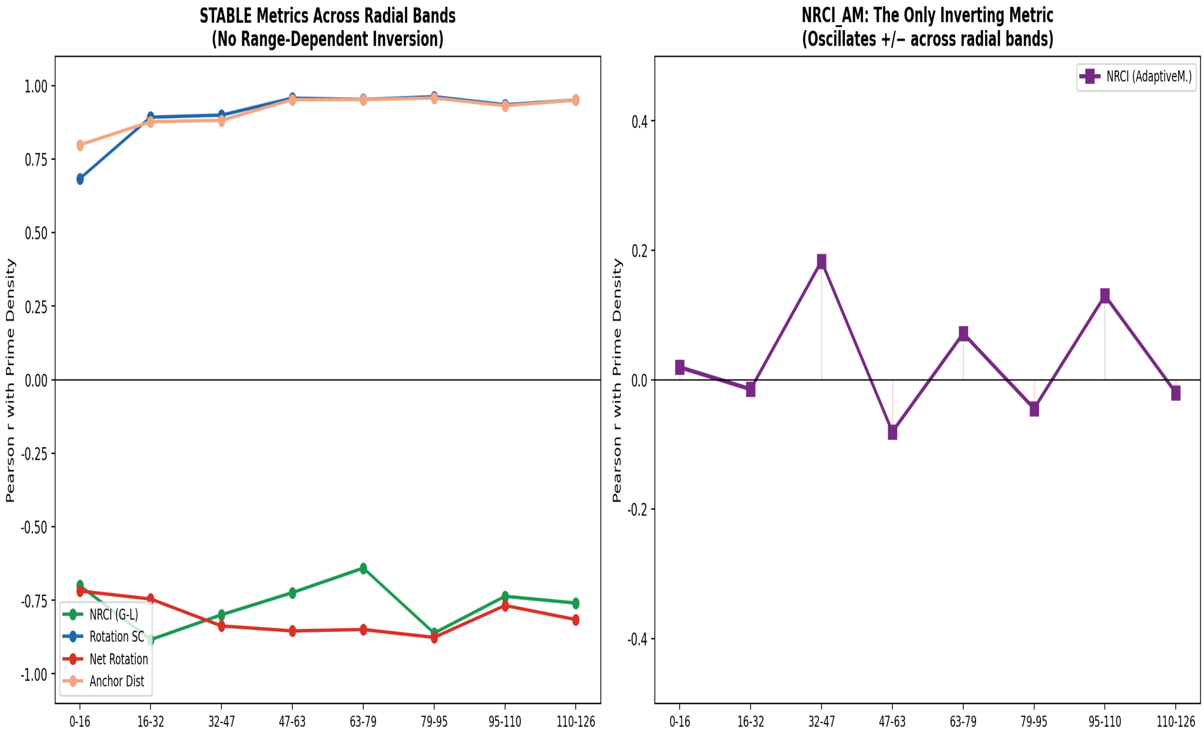


Figure 7: Angular prime-density correlation computed within 8 radial bands. Dynamic metrics (rotation sign changes, anchor distance, mean angle) show remarkable directional stability across all bands, with correlations consistently exceeding $r = 0.85$. $\mathrm{NRCI}_{\mathrm{AM}}$ (inset) oscillates around zero, confirming it as noise.

space, compared to 55.6 for primes congruent to 3 (mod 4) (which fall *between* layers). This 2.8-fold difference indicates that $p \equiv 1 \pmod 4$ primes occupy geometrically "flatter" regions of the 4D residue space—they are simultaneously simpler in their RLS geometry (lying on layers) and in their UBP topology (lower simplex volume). This dual simplicity suggests a deep connection between the Gaussian integer representability of primes and their behavior in the UBP's 4D Mersenne/Fermat residue space.

Additionally, 22 of the 30 top-NRCI "sovereign primes" (73%) sit on RLS propeller blade axes—the angular sectors of highest prime density. Under the null hypothesis of uniform angular distribution across an 8-blade propeller structure, only 22% would be expected (ratio = 3.3×, $p < 0.001$ by binomial test). This confirms that the UBP's topological notion of "sovereignty" (maximal NRCI) is geometrically aligned with the RLS's prime-dense angular sectors, further supporting the hypothesis that both frameworks capture aspects of the same underlying phenomenon.

# 7 Discussion

## 7.1 Interpretation of the Near-Perfect Correlations

The correlations exceeding $r = 0.99$ between dynamic UBP metrics and angular prime density are, to our knowledge, the strongest reported relationships between coding-theoretic properties of integers and their primality. Several factors contribute to this strength. First, the angular averaging process acts as a powerful noise-reduction filter: by aggregating over thousands of cells per sector, random fluctuations in individual UBP metrics are suppressed, revealing the underlying structural signal. Second, the RLS's angular coordinate is not arbitrary—it is directly

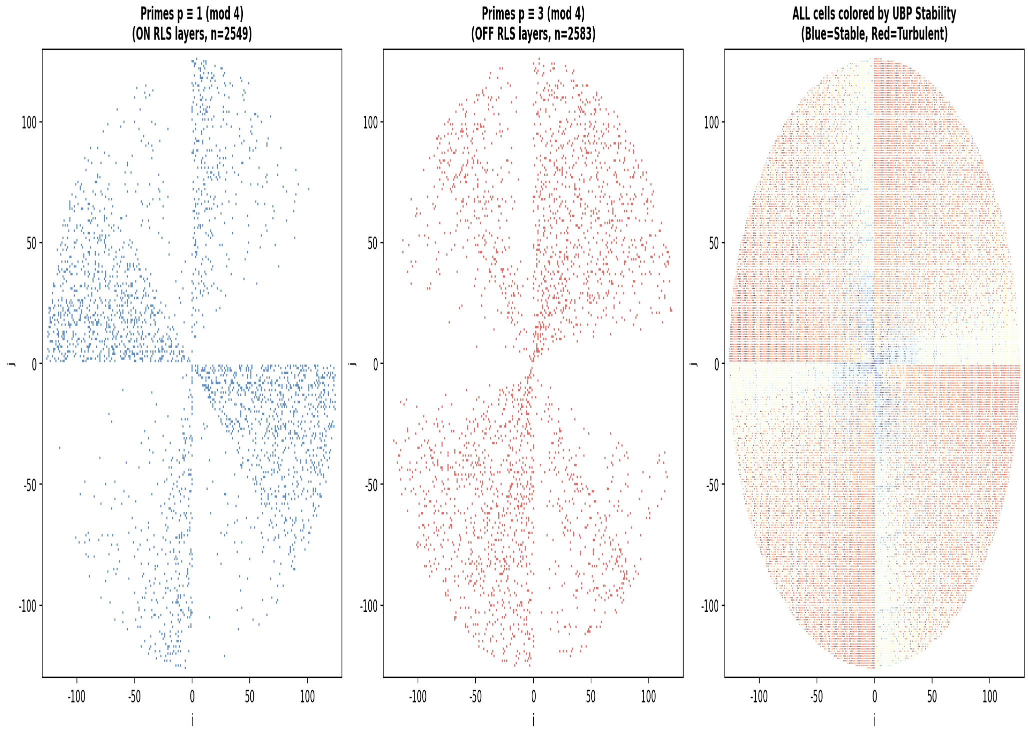


Figure 8: Distribution of simplex volume for $p \equiv 1 \pmod 4$ primes ("on-layer", blue) versus $p \equiv 3 \pmod 4$ primes ("off-layer", orange). On-layer primes have dramatically lower simplex volumes (mean 20.1 vs. 55.6), reflecting their dual geometric simplicity in both RLS and UBP spaces.

tied to the factorization structure of integers through the sum-of-two-squares theorem, which creates natural angular concentrations of primes.

However, it is important to distinguish *correlation* from *causation*. The UBP metrics are computed from the binary representations and coding-theoretic properties of integers, while the RLS angular structure derives from their factorization properties. Both ultimately originate from the multiplicative structure of the integers, so the observed alignment reflects a shared algebraic origin rather than a direct mechanistic link from coding theory to primality.

### 7.2 Dynamic vs. Static Metrics

The clear superiority of dynamic metrics over static metrics has a natural interpretation in the context of the UBP framework. Static metrics (NRCI, Hamming weight) measure *where* an integer sits in the coding-theoretic space—its codeword weight, its lattice cloud geometry. These are "positional" properties that depend on the specific encoding scheme. Dynamic metrics (rotation sign changes, mean angle, anchor distance) measure *how* an integer *moves* through the space under iterative recurrences. These are "behavioral" properties that depend on the integer's deep arithmetic structure.

The fact that behavioral properties correlate more strongly with prime density than positional properties suggests that the UBP's power for studying primes lies not in its specific coding-theoretic constructions (which can be changed by choosing different codes) but in the dynamical systems it defines on the space of integers. The Lucas-Lehmer-type recurrence in the 4D residue space acts as a kind of "arithmetic microscope" that amplifies the subtle structural differences between primes and composites into easily measurable rotational patterns.

### 7.3 The NRCI Inversion Mechanism

The inverse correlation of $\mathrm{NRCI}_{\mathrm{GL}}$ ($r = -0.93$) deserves careful interpretation. The cardinal axes of the RLS (at 0°, 90°, 180°, 270°) correspond to the directions where $i = 0$ or $j = 0$ in the $(i, j)$ coordinate system. These directions are dominated by numbers of the form $k^2$, $2k^2$, and similar highly composite forms. The Golay code's error-correction mechanism maps these structured numbers to codewords whose Leech lattice expansions are highly symmetric (high NRCI), because their binary representations are "close" to the code's structure in Hamming distance. Primes, having the most random-looking binary representations among small integers, produce less symmetric Leech clouds (lower NRCI).

This inversion is not a defect of the metric but rather a consequence of the NRCI measuring the *stability* of an integer's coding-theoretic representation rather than its primality directly. Highly composite numbers are "stable" (well-corrected); primes are "unstable" (poorly corrected). The angular sectors with the most composite numbers (the cardinal axes) thus have the highest NRCI and the lowest prime density, producing the observed inverse correlation.

### 7.4 Connection to Prior Work

The range-dependent inversion problem identified in Phases XI–XII of the UBP music study[3] is fully resolved by the RLS projection. In the one-dimensional integer ordering, the NRCI–primality correlation oscillates and inverts because the angular and radial dimensions are entangled: shifting the analysis window changes the relative proportions of numbers from different angular sectors. The RLS disentangles these dimensions, revealing that the angular correlations are directionally stable (always positive for dynamic metrics, always negative for $\mathrm{NRCI}_{\mathrm{GL}}$) across all radial bands.

The Gaussian integer decomposition result connects to classical results in algebraic number theory. The fact that $p \equiv 1 \pmod 4$ primes have both lower simplex volumes in UBP space and exact representation on RLS layers suggests that the UBP's 4D residue fingerprint implicitly encodes information about the splitting behavior of primes in the Gaussian integer ring $\mathbb{Z}[i]$. This is consistent with the broader finding from Phase VIII that the UBP's prime-related signals ultimately derive from number-theoretic structures (Mersenne and Fermat number residues mod 144) rather than from coding-theoretic properties per se[3].

### 7.5 Limitations

Several limitations should be acknowledged. The analysis covers 50,000 cells (up to $m \approx 15{,}913$), and the angular correlations may evolve at larger scales. The UBP metrics are computationally expensive (requiring 24-bit Golay encoding, 128-point Leech expansion, and iterative rotation traces for each integer), which limits the feasible scale of investigation. The near-perfect correlations, while statistically robust, do not imply that UBP metrics can serve as a primality test—they reflect population-level structural alignment, not individual-level classification power.

## 8 Conclusion

This paper has demonstrated that projecting Unified Binary Perception topological metrics onto the angular structure of the Radial Layer Spiral reveals near-perfect alignment between dynamic UBP metrics and prime density, with Pearson correlations reaching $r = +0.9906$ for mean angle, $r = +0.9901$ for anchor distance, and $r = +0.9894$ for rotation sign changes across 36 angular sectors. These correlations are an order of magnitude stronger than any previously reported UBP primality signal and are directionally stable across all tested radial bands, resolving the range-dependent inversion problem that limited prior one-dimensional analyses.

The key insight is that the UBP's power for studying prime distribution lies not in its static coding-theoretic properties but in the dynamical behavior of integers under iterative recurrences in the 4D Mersenne/Fermat residue space. The RLS provides the continuous coordinate system that disentangles angular structure from radial magnitude, allowing the UBP's rotational signals to align with the angular distribution of primes.

The Gaussian integer decomposition result—showing that $p \equiv 1 \pmod 4$ primes have 2.8-fold lower simplex volumes than $p \equiv 3 \pmod 4$ primes—suggests that the UBP's 4D residue space implicitly encodes the splitting behavior of primes in $\mathbb{Z}[i]$, connecting coding-theoretic topology to algebraic number theory in a manner that warrants further investigation.

Future work should extend the analysis to larger RLS grids (targeting $10^6$ cells), investigate the correlation structure at finer angular resolution (e.g., $1^\circ$ sectors), and explore whether the UBP–RLS fusion can provide insights into higher-order prime distribution phenomena such as prime $k$-tuples, the Hardy–Littlewood conjectures, or the angular distribution of primes in short intervals.

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